

Research, engineering and team leadership.

I am an Applied Scientist for Digital Biology at NVIDIA and a PhD candidate at Oxford.
My earlier roles include global life-sciences research alliances, research and engineering management, production machine learning and product work.

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Scope

Research, systems and teams

My experience includes first-author research, production systems, team management and work with external organisations.

8 researchers and engineers managed	100+ AI startups supported on GPU workflows	5,000+ developers reached through workshops, symposiums and training	55% lower cPCV in an A/B- tested ML system
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Selected experience

Experience

2024–present	Applied Scientist for Digital Biology · NVIDIA Research on generative models, geometric learning and biomolecular machine learning. First author of Neural Point-Forms and Entropy Across the Bridge; contributor to BioNeMo Framework examples and multi-node workflows. - Derived a conditional–marginal entropy-rate scheduler with an 18.1% MMD improvement over linear scheduling in a 10-step ODE-Heun evaluation. - Developed a permutation-invariant point-cloud layer based on diffusion geometry and form-comparison matrices.
2023–2024	Global Life Sciences Research Alliances Lead · NVIDIA Led research collaborations with universities, startups and life-sciences organisations working on biomolecular models and GPU systems. - Managed an eight-person research and engineering team on peptide inverse folding; the resulting method increased sequence diversity by 20% while maintaining structural similarity. - Coordinated work on model evaluation, use of Cambridge-1, multi-node training and scientific workshops.
2022	Developer Relations Manager, Healthcare & Life Sciences Startups Lead EMEA · NVIDIA Worked with founders and technical teams on GPU-accelerated scientific machine learning. Helped organise workshops and developer programmes that reached more than 2,000 people.

2021–2022	Senior Data Scientist · GSK Built and deployed a reinforcement-learning and Bayesian system for multi-objective campaign allocation. The system reduced cPCV by 55% in an A/B test against two large platform providers. - Built the Spark and Databricks data pipelines, PyTorch models, APIs, Azure deployment and web interface. - Mentored five apprentices, directly managed two and taught Bayesian and explainable-ML methods to more than 100 colleagues.
2020–2021	Product Manager, AI & Analytics · GSK Led three data engineers working on EMEA data pipelines, forecasting and analytics tools. Agreed project goals and measures with regional teams.
2018–2019	AI Startups Ecosystem Manager · NVIDIA Advised more than 100 AI startups on GPU-accelerated ML design, RAPIDS and NVIDIA SDKs through technical sessions, workshops and founder programmes.
2008–2020	Earlier engineering, analytics and product roles Machine-learning research placement at KIT-AR/UCL; commercial analytics at Hotelbeds; data and analytics consulting at Alliantis; innovation engineering at GVT; IT audit at EY; and mainframe software engineering at HSBC.

Education

Degrees and doctoral work

My doctoral work is supervised by Professor Michael Bronstein.

2024–present	PhD in Computer Science · University of Oxford Generative models, geometric deep learning and AI for science.
2019–2020	MSc Data Science & Machine Learning · University College London Distinction. Deep learning, reinforcement learning, probabilistic modelling and graphical models.
2008–2013	BSc Information Systems; postgraduate diploma in Telematics & Computer Networks Pontifical Catholic University of Paraná and Federal University of Technology – Paraná. Selected by PUCPR, on the basis of strong academic performance, for a cultural exchange programme at Université de Technologie de Compiègne, France, in 2009; awarded a CREPUQ scholarship for a semester at Bishop’s University, Canada, in 2011.

Recognition and credentials					
Awards, selections and professional credentials					
2024	Creator Fund Student Investment Partner Selected for the 2024–25 cohort of seven Student Investment Partners from across Europe.	2022	GSK Gold Award — OneID Team award recognising OneID’s global impact and work that significantly improved data-science capabilities across field operations.	2022	GSK Bronze Award — Haleon demerger Recognition for contributing to the demerger of Haleon from GSK.
2022	DataIQ Industry Awards shortlist OneID was shortlisted in the top three for the “Breakthrough with Data” category.	Dec 2019	Best Business Solution DSS Microsoft/Amex Hackathon, UCL Data Science.	2017	British Computer Society Awarded professional membership.
2012	ITIL V3 Foundation EXIN certification.	2009	Second place — III Mobile Games Workshop PUCPR recognition for the game BugBusters.		